

Lecture 3: Text Classification (I)

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CSCI 680: Natural Language Processing
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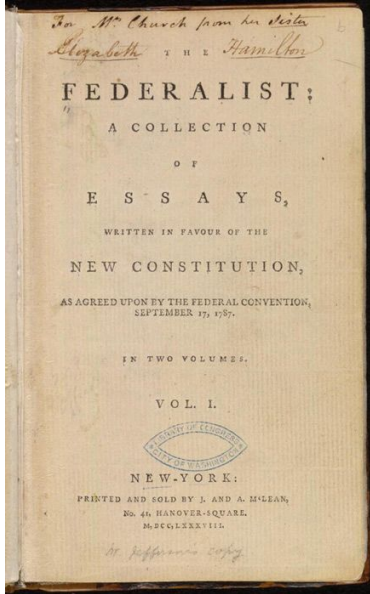
Many slides are adapted from Jurafsky & Martin (3rd ed.)

Lecture Outline

- The Task of Text Classification
- Naive Bayes
- Logistic Regression

The Task of Text Classification

Text Classification



Authorship
Identification



Sentiment analysis

Google Translate

Text

Im

Detect language



Language
identification

Assigning
subject
categories,
topics, or
genres,
Spam
detection,
Etc.

Text Classification: Definition

Input:

- a "document" (which can be any text) d
- a fixed set of classes $Y = \{y_1, y_2, \dots, y_J\}$

Output: a predicted class $\hat{y} \in Y$

The hat or circumflex notation $\hat{y}$ is used to refer to an estimated or predicted value

Most Common Classification Method: Supervised Machine Learning

Training:

Input:

- a fixed set of classes $Y = \{y_1, y_2, \dots, y_J\}$
- a training set of m hand-labeled inputs $(x^{(1)}, y^{(1)}), \dots, (x^{(m)}, y^{(m)})$

Output:

- a learned classifier $\gamma: d \rightarrow \hat{y}$

Inference or Test

Input: a document d

Output: a class $\hat{y}$

Supervised Machine Learning for Classification

Many kinds of classifiers!

- Logistic regression
- Naive Bayes
- Neural networks
- k-nearest neighbors
- LLMs
- Fine-tuned as classifiers
- Prompted to give a classification

Naive Bayes Classification

Naive Bayes Intuition

- Simple ("naive") classification method based on Bayes rule
- Relies on very simple representation of document: **Bag of words**

The Bag-of-Words Representation

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

The Bag-of-Words Representation

γ (

seen	2
sweet	1
whimsical	1
recommend	1
happy	1
...	...

) = γ




Bayes' Rule Applied to Documents and Classes

- For a document d and a class y

posterior

likelihood

prior

$$P(y \mid d) = \frac{P(d \mid y)P(y)}{P(d)}$$

marginal

Naive Bayes Classifier (I)

$$y_{MAP} = \arg \max_{y \in Y} P(y \mid d)$$

MAP is “maximum a posteriori” = most likely class

$$y_{MAP} = \arg \max_{y \in Y} \frac{P(d \mid y)P(y)}{P(d)}$$

Bayes Rule

$$y_{MAP} = \arg \max_{y \in Y} P(d \mid y)P(y)$$

Dropping the denominator (marginal is the same across classes)

Naive Bayes Classifier (II)

likelihood

prior

$$y_{MAP} = \arg \max_{y \in Y} P(d \mid y) P(y)$$

$$\arg \max_{y \in Y} P(x_1, x_2, \dots, x_n \mid y) P(y)$$

Document d
represented as
features $x_1, \dots, x_n$

Multinomial Naive Bayes Independence Assumptions

$$P(x_1, x_2, \dots, x_n \mid y)$$

Bag-of-Words Assumption: Assume that position doesn't matter

Conditional Independence: Assume the feature probabilities $P(x_i | y_j)$ are independent given the class y .

$$P(x_1, x_2, \dots, x_n \mid y) = P(x_1 \mid y)P(x_2 \mid y) \cdots P(x_n \mid y)$$

Multinomial Naive Bayes Classifier

$$y_{MAP} = \arg \max_{y \in Y} P(x_1, x_2, \dots, x_n \mid y) P(y)$$

$$y_{NB} = \arg \max_{y_j \in Y} P(y_j) \prod_{i=1}^n P(x_i \mid y_j)$$

Applying Multinomial Naive Bayes Classifiers to Text Classification

positions $\leftarrow$ all word positions in test document

$$y_{NB} = \arg \max_{y_j \in Y} P(y_j) \prod_{i \in \text{positions}} P(x_i \mid y_j)$$



$$y_{NB} = \arg \max_{y_j \in Y} \left[\log P(y_j) + \sum_{i \in \text{positions}} \log P(x_i \mid y_j) \right]$$

Learning the Multinomial Naive Bayes Model

Use frequencies in the data:

$$\hat{P}(y_j) = \frac{N_{y_j}}{N_{total}}$$

Documents belonging
to class y_j

$$\hat{P}(w_i \mid y_j) = \frac{\text{count}(w_i, y_j)}{\sum_{w \in V} \text{count}(w, y_j)}$$

fraction of times word
 w_i appears
among all words in
documents of topic y_j

- Create mega-document for topic j by concatenating all docs in this topic
 - Use frequency of w in mega-document

Laplace (add-1) smoothing for Naive Bayes

With smoothing!

$$\begin{aligned}\hat{P}(w_i \mid y) &= \frac{\text{count}(w_i, y) + 1}{\sum_{w \in V} (\text{count}(w, y) + 1)} \\ &= \frac{\text{count}(w_i, y) + 1}{(\sum_{w \in V} \text{count}(w, y)) + |V|}\end{aligned}$$

Multinomial Naive Bayes: Learning

From training corpus, extract *Vocabulary* V

Calculate $P(y_j)$ terms:

- For each y_j in Y do
 $docs_j \leftarrow$ all docs with class y_j

$$P(y_j) \leftarrow \frac{|docs_j|}{|\text{total \# documents}|}$$

Calculate $P(w_k | y_j)$ terms:

- $Text_j \leftarrow$ single doc containing all $docs_j$
- For each word w_k in *Vocabulary*
 $n_k \leftarrow$ # of occurrences of w_k in $Text_j$

$$P(w_k | y_j) \leftarrow \frac{n_k + \alpha}{n + \alpha |Vocabulary|}$$

Unknown Words and Stop Words

- What about **unknown words** that appear in our test data but not in our training data or vocabulary?
 - We ignore them: remove them from the test document!
 - Don't include any probability for them at all
- Some systems ignore **stop words**
 - Stop words: very frequent words like the and a.
 - Sort the vocabulary by word frequency in training set
 - Call the top 10 or 50 words the stopword list.
 - Remove all stop words from both training and test sets

Sentiment Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

Prior from training

Likelihoods from training

Scoring the test set

$$\hat{P}(y_j) = \frac{N_{y_j}}{N_{total}} = \frac{\text{count}(w_i, y) + 1}{(\sum_{w \in V} \text{count}(w, y)) + |V|} \quad \arg \max_{y_j \in Y} \left[\log P(y_j) + \sum_{i \in \text{positions}} \log P(x_i | y_j) \right]$$

Logistic Regression Classification

Components of a probabilistic machine learning classifier

Given m input/output pairs $(x^{(i)}, y^{(i)})$:

1. A **feature representation** of the input. For each input observation $x^{(i)}$, a vector of features $[x_1, x_2, \dots, x_n]$. Feature j for input $x^{(i)}$ is x_j , more completely $x_j^{(i)}$, or sometimes $f_j(x)$.
2. A **classification function** that computes $\hat{y}$, the estimated class, via $p(y|x)$, like the **sigmoid** or **softmax** functions.
3. An objective function for learning, like **cross-entropy loss**.
4. An algorithm for optimizing the objective function: **stochastic gradient descent**.

Logistic Regression

- Important analytic tool in natural and social sciences
- Baseline supervised machine learning tool for classification
- Is also the foundation of neural networks

The Two Phases of Logistic Regression

Training (next lecture): we learn weights w and b using **stochastic gradient descent** and **cross-entropy loss**.

Test (also called **inference**) (this lecture): Given a test example x we compute $p(y|x)$ using learned weights w and b , and return whichever label ($y=1$ or $y=0$) has higher probability

Binary Classification in Logistic Regression

Given a series of input/output pairs: $(x^{(i)}, y^{(i)})$

For each observation $x^{(i)}$:

- We represent by a **feature vector** $[x_1, x_2, \dots, x_n]$
- We compute an output: a predicted class $\hat{y}^{(i)} \in \{0, 1\}$

Features in Logistic Regression

- For feature x_i , weight w_i tells is how important is x_i
 - x_i = "review contains 'awesome'": $w_i = +10$
 - x_j = "review contains 'abysmal'": $w_j = -10$
 - x_k = "review contains 'mediocre'": $w_k = -2$

Logistic Regression for One Observation x

- Input observation: vector $x = [x_1, x_2, \dots, x_n]$
 - Weights: one per feature: $W = [w_1, w_2, \dots, w_n]$
 - Sometimes we call the weights $\theta = [\theta_1, \theta_2, \dots, \theta_n]$
 - Output: a predicted class $\hat{y}^{(i)} \in \{0, 1\}$
-
- (multinomial logistic regression: $\hat{y}^{(i)} \in \{0, 1, 2, 3, 4\}$)

How to do Classification

- For each feature x_i , weight w_i tells us the importance of x_i
 - Plus we'll have a bias b
- We'll sum up all the weighted features and the bias

$$z = \left(\sum_{i=1}^n w_i x_i \right) + b$$
$$z = w \cdot x + b$$

- If this sum is high, we say $y=1$; if low, then $y=0$

But We Want a Probabilistic Classifier

- We need to formalize “sum is high”.
- We’d like a principled classifier that gives us a probability, just like Naive Bayes did
- We want a model that can tell us:
 - $p(y=1|x; \theta)$
 - $p(y=0|x; \theta)$

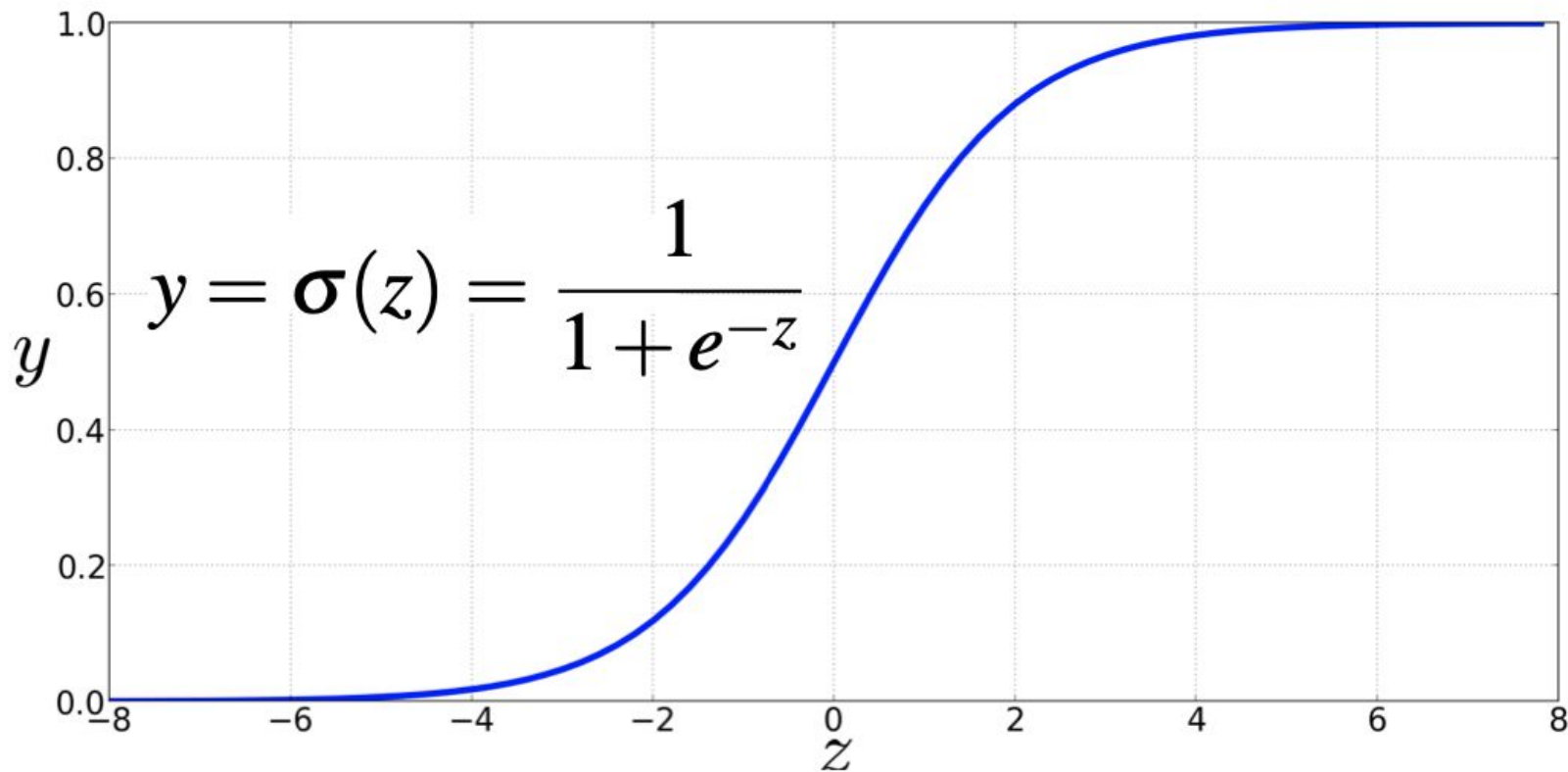
The Problem: z Isn't a Probability, It's Just A Number

$$z = w \cdot x + b$$

Solution: use a function of z that goes from 0 to 1

$$y = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + \exp(-z)}$$

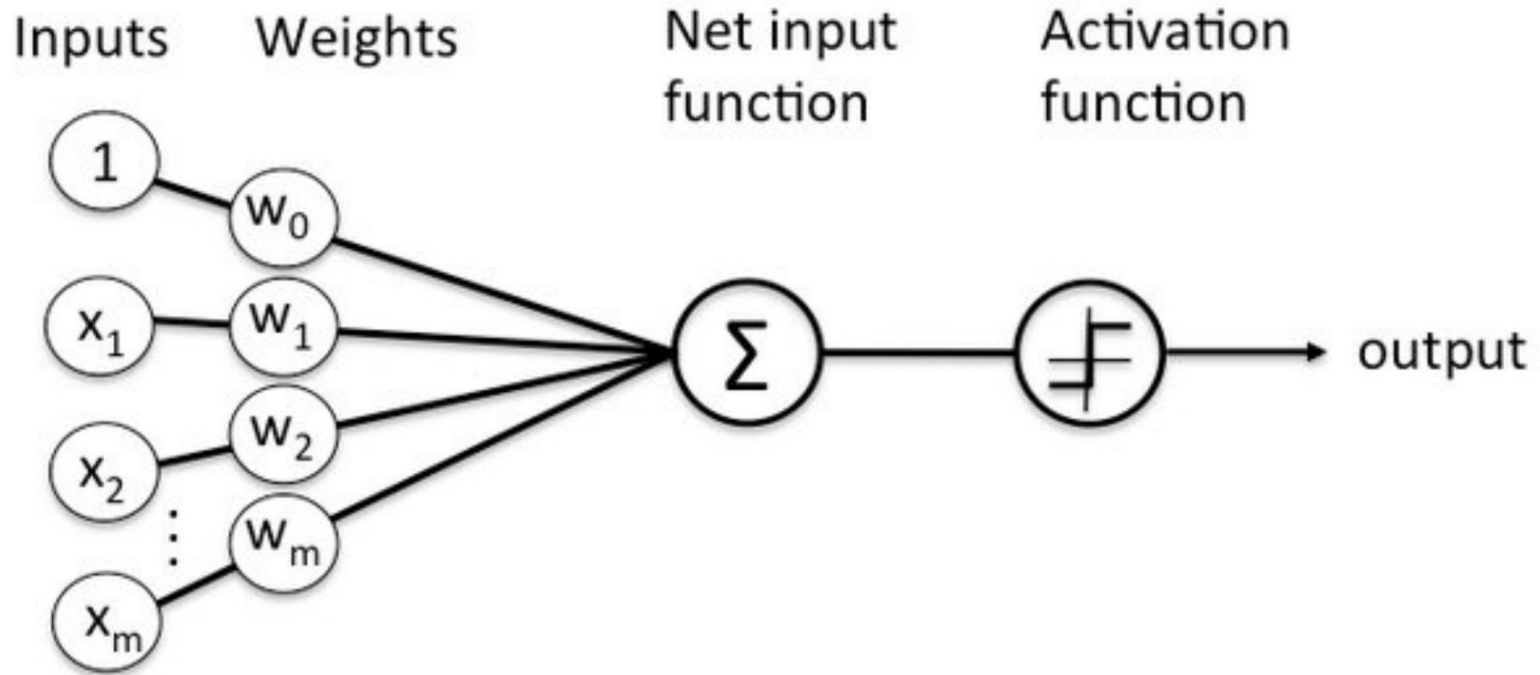
The Very Useful Sigmoid or Logistic Function



Idea of Logistic Regression

- We'll compute $w \cdot x + b$
- And then we'll pass it through the sigmoid function:
$$\sigma(w \cdot x + b)$$
- And we'll just treat it as a probability

Another Notation



Making Probabilities with Sigmoids

$$\begin{aligned} P(y = 1) &= \sigma(w \cdot x + b) \\ &= \frac{1}{1 + \exp(-(w \cdot x + b))} \end{aligned}$$

$$\begin{aligned} P(y = 0) &= 1 - \sigma(w \cdot x + b) \\ &= 1 - \frac{1}{1 + \exp(-(w \cdot x + b))} \\ &= \frac{\exp(-(w \cdot x + b))}{1 + \exp(-(w \cdot x + b))} \end{aligned}$$

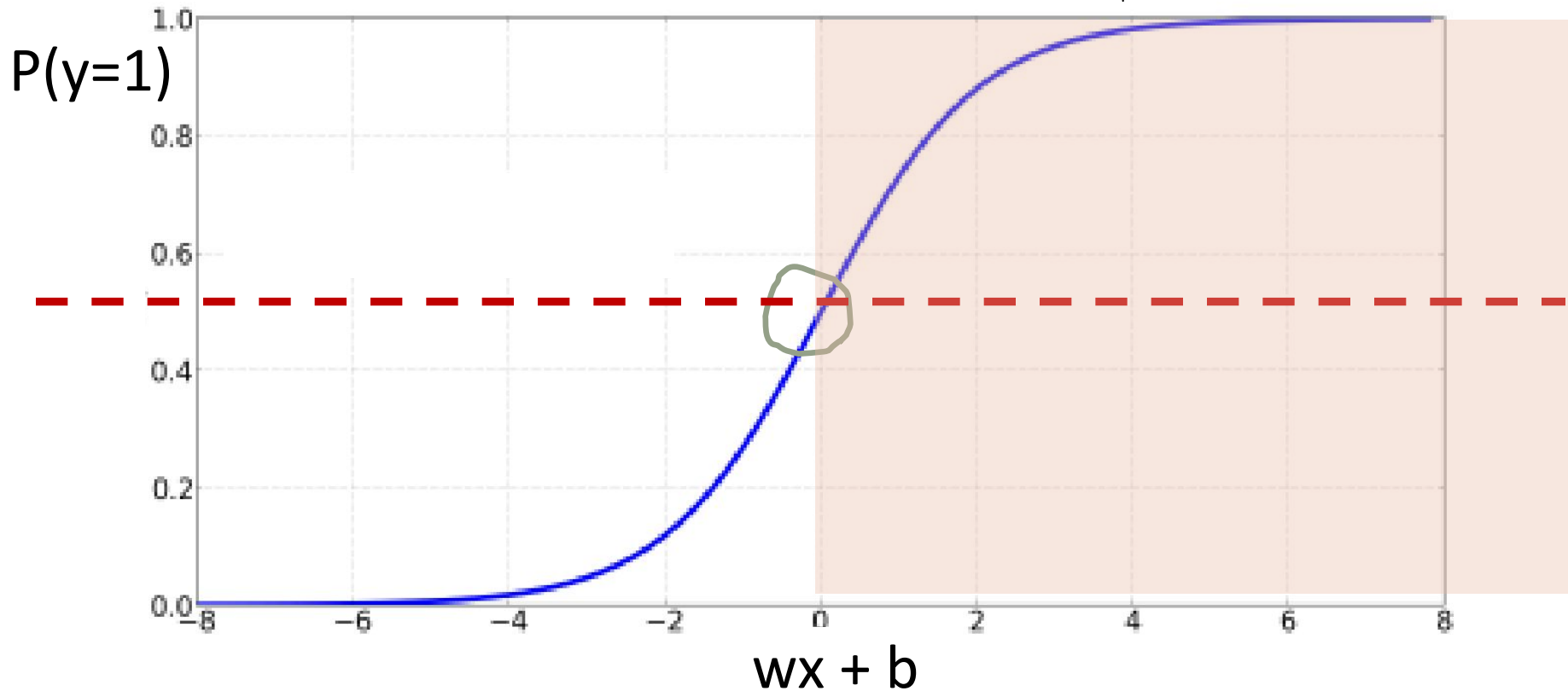
Turning a Probability into a Classifier

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|x) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

0.5 here is called the decision boundary

The Probabilistic Classifier

$$P(y=1) = \sigma(w \cdot x + b)$$
$$= \frac{1}{1 + e^{-(w \cdot x + b)}}$$



Turning a Probability into a Classifier

$$\hat{y} = \begin{cases} 1 & \text{if } P(y = 1|x) > 0.5 \\ 0 & \text{otherwise} \end{cases} \quad \begin{array}{l} \text{if } w \cdot x + b > 0 \\ \text{if } w \cdot x + b \leq 0 \end{array}$$

Logistic Regression: A Text Example on Sentiment Classification

It's hokey . There are virtually no surprises , and the writing is second-rate .
So why was it so enjoyable ? For one thing , the cast is
great . Another nice touch is the music . I was overcome with the urge to get
off the couch and start dancing . It sucked me in , and it'll do the same to you

It's **hokey**. There are virtually **no** surprises, and the writing is **second-rate**. So why was it so **enjoyable**? For one thing, the cast is **great**. Another **nice** touch is the music. **I** was overcome with the urge to get off the couch and start dancing. It sucked **me** in, and it'll do the same to **you**.

Diagram illustrating feature extraction from the text:

- $x_2 = 2$ (connected to "no")
- $x_3 = 1$ (connected to "I")
- $x_1 = 3$ (connected to "great", "nice", "me")
- $x_5 = 0$ (connected to "!", "I")
- $x_6 = 4.19$ (connected to "great", "nice", "me", "you")
- $x_4 = 3$ (connected to "I", "me", "you")

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no" } \in \text{ doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!" } \in \text{ doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(66) = 4.19$

Classifying sentiment for input x

Var	Definition	Va	5.2
x_1	count(positive lexicon) $\in$ doc)	3	
x_2	count(negative lexicon) $\in$ doc)	2	
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1	
x_4	count(1st and 2nd pronouns $\in$ doc)	3	
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0	
x_6	log(word count of doc)	$\ln(66) = 4.19$	

Suppose $w = [2.5, -5.0, -1.2, 0.5, 2.0, 0.7]$

$$b = 0.1$$

Classifying sentiment for input x

$$\begin{aligned} p(+|x) = P(Y = 1|x) &= \sigma(w \cdot x + b) \\ &= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.19] + 0.1) \\ &= \sigma(.833) \\ &= 0.70 \end{aligned}$$

$$\begin{aligned} p(-|x) = P(Y = 0|x) &= 1 - \sigma(w \cdot x + b) \\ &= 0.30 \end{aligned}$$

We can build features for logistic regression for any classification task: period disambiguation

This ends in a period. End of sentence
The house at 465 Main St. is new. Not end

$$x_1 = \begin{cases} 1 & \text{if } \textit{Case}(w_i) = \textit{Lower} \\ 0 & \text{otherwise} \end{cases}$$

$$x_2 = \begin{cases} 1 & \text{if } w_i \in \textit{AcronymDict} \\ 0 & \text{otherwise} \end{cases}$$

$$x_3 = \begin{cases} 1 & \text{if } w_i = \textit{St.} \ \& \ \textit{Case}(w_{i-1}) = \textit{Cap} \\ 0 & \text{otherwise} \end{cases}$$

Classification in (binary) logistic regression: summary

Given:

- a set of classes: (+ sentiment, - sentiment)
- a vector $\mathbf{x}$ of features $[x_1, x_2, \dots, x_n]$
 - $x_1 = \text{count}(\text{"awesome"})$
 - $x_2 = \log(\text{number of words in review})$
- A vector $\mathbf{w}$ of weights $[w_1, w_2, \dots, w_n]$
 - w_i for each feature f_i

$$\begin{aligned} P(y = 1) &= \sigma(w \cdot x + b) \\ &= \frac{1}{1 + e^{-(w \cdot x + b)}} \end{aligned}$$

TODOs:

- **Continue brainstorming for your project.**

Next Lecture: Text Classification (II)